

Internship Task Report: Task 9 Day 9

Date: [16.06.2026]

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(Interns Batch):- Cyber security

Target Website: <https://kryosync.netlify.app>

Task:-Report on types of cyber Attacks.

Part 1: Analysis of Critical Cyber Attacks

1. Phishing

Phishing is a form of cyber attack where malicious actors disguise themselves as trustworthy entities to trick individuals into revealing sensitive information, such as login credentials, financial data, or personally identifiable information (PII).

How it Works: Attackers send fraudulent communications (emails, SMS, or direct messages) containing links to spoofed websites. When a user logs in or inputs data on the fraudulent site, the attacker harvests the credentials.

Variations: * Spear Phishing: Highly targeted attacks aimed at specific individuals or organizations using gathered personal details.

Whaling: Spear phishing specifically directed at high-profile executives (CEOs, CFOs).

Clone Phishing: Copying a legitimate, previously delivered email and swapping the safe link or attachment with malicious one.

2. Malware

Malware (Malicious Software) is a broad term encompassing any software intentionally designed to disrupt, damage, or gain unauthorized access to a computer system, network, or server.

Delivery Mechanisms: It is often delivered via malicious email attachments, infected software downloads, drive-by downloads (visiting a compromised website), or removable media.

Common Types:

Viruses and Worms: Self-replicating programs that spread across systems and networks.

Trojan Horses: Programs disguised as legitimate software that open backdoors for attackers once executed.

Spyware/Keyloggers: Software that secretly monitors user activity and captures keystrokes to steal sensitive data.

3. Ransomware

Ransomware is a specialized, highly destructive subset of malware that encrypts a victim's files, rendering the system or data inaccessible. The attackers then demand a ransom payment (usually in cryptocurrency) in exchange for the decryption key.

The Anatomy of an Attack: Once executed inside a network, ransomware moves laterally to infect as many endpoints and backup servers as possible before executing the encryption routine.

Double Extortion: Modern ransomware tactics involve exfiltrating (stealing) sensitive data before encrypting it. Attackers threaten to leak or sell the data publicly if the ransom is not paid, bypassing the safety net of traditional offline backups.

4. Social Engineering

Social engineering is the psychological manipulation of people into performing actions or divulging confidential information. Unlike technical hacks that exploit software vulnerabilities, social engineering exploits human psychology (trust, fear, urgency, or curiosity).

Core Concepts: Technology-based attacks often fail against strong firewalls, making the human element the weakest link in any security perimeter.

Common Tactics:

Pretexting: Fabricating a scenario (e.g., pretending to be an IT support technician or an auditor) to manipulate a victim into granting access.

Baiting: Offering an item or goodie (like a free software download or an infected USB drive left in a public space) to entice victims.

Quid Pro Quo: Promising a service or benefit in exchange for information (e.g., "I can fix your computer lag if you disable your antivirus").

Part 2: Vulnerability Mapping & Prevention Techniques for Kryosync

Kryosync operates as a digital agency specializing in Web Development, Software Engineering, UI/UX Design, and IT Consulting. As a business built entirely around deploying digital products for clients, safeguarding both its public-facing portal (hosted on Netlify) and its back-end workflows is paramount.

Below are the mapped prevention techniques required to insulate Kryosync from the discussed cyber threats.

1. Website & Infrastructure Security (Netlify Specifics)

Because the platform is deployed via Netlify, it leverages a Jamstack/static architecture. This inherently removes many traditional server-side vulnerabilities (like SQL injections on the front-end), but specific security controls must still be enforced: